UAES	订 货 规 程 **Order Specification**	编号：F 01R L0Y 033 No.: F 01R L0Y 033 版本：02 Version: 编制日期：220505 Issued Date: 页数：第1页 共 17 页 Page of

产品/Product: 节气门壳体 Throttle Housing

型号/Type:

零件编号/Part Number: F 01R 10Y 108/…

属/Belonging to: F 01R 00Y 094/…

序号 No	更改页 Page	修订 Revision	日期 Date	编制 Draftsman	校对 Check	审核 Review	标准化 Standardi-zation	批准 Approval
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目录

1 综述
1.1 包装和运输
1.2 零件识别
2 机加工制造要求
3 附加生产物料
4 清洗过程要求
5 机加壳体的终检验
5.1 目检
5.1.1 毛刺
5.1.2 毛刺和废屑
5.1.3 气孔和缩孔
5.2 水管处的泄露测试（如果有水管）
6 允许的热裂纹
7 定义 H 面的测量点
8 定义铝壳体密封面的平面度测量
9 毛坯壳体的基准及方向
10 机加工壳体的基准及方向

List of Contents

1 Overview
1.1 Packing and Delivery
1.2 Identification Of The Parts
2 Manufacturing Demands On Machining
3 Additional Manufacturing Substance
4 Manufacturing Demands On Washing Process
5 Final Tests On Machined Throttle Housing
5.1 Visual Tests
5.1.1 Burrs
5.1.2 Borings
5.1.3 Pores and Shrink holes
5.2 Leakage Test Water Heating (If Existing)
6 Permissible Heat Cracks
7 Measuring-points to define the plane H
8 Definition flatness measurement Throttle Housing (cover sealing surface)
9 Orientation of the not machined throttle housing
10 Orientation of the machined throttle housing

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UAES	订 货 规 程 **Order Specification**	编号：F 01R L0Y 033 No.: F 01R L0Y 033 版本：02 Version: 编制日期：220505 Issued Date: 页数：第3页 共 17 页 Page of

1 综述

压铸铝壳体是节气门总成的一部分。车辆的驾驶性和驾驶安全与节气门体的功能和可靠性直接相关。因此在生产，运输和测试过程中的清洁度很关键。有损伤，或者状态不明的零件不得使用。本订货规程中的图片供参考，根据不同型号的壳体，所指示的区域会略有差异。

1.1 包装和运输

详细的包装和运输规范见 XaP 的具体要求。

1.2 零件识别

零件必须保证最小批次的可追溯性，同时不同的供应商可识别。

2 机加工制造要求

过程控制

设备和工艺过程的稳定性须进行统计和计算。必须保证图纸中所标注相关尺寸的过程能力。相关尺寸(Cpk≥1.33)在生产过程中必须监控并进行评价。相关尺寸需要使用过程控制的表格进行记录，并随时可查。数据保存至少三年。

3 附加生产物料

以下所涉及的润滑和冷却液无须额外认可：

油: ARAL Sulnit HM – 见规格 3170, 状态 03.12.99

乳状液: ARAL Sarol 471 EP ca 15%自来水, 状态 01.12.99

Castrol Alusol XT FF, 状态 12.02.2009

1 Overview

The die casted housing of the throttle device is part of the ETC. The drivability and the driving safety of the car depend on the function and the reliability of the ETC system. That‘s why biggest care and cleanliness are absolute necessary during manufacturing, handling and testing. Damaged parts and parts which may not guarantee an impeccable status must not be used any longer. All shown pictures in that document are symbolical. According to the special type of the throttle housing the shown areas can differ.

1.1 Packaging and Delivery

The demands for delivery and packaging are fixed separately in a packaging instruction by XaP.

1.2 Identification of the Parts

Traceability of the parts must be guaranteed up to the smallest batch. The supplier has to be identifiable.

2. Manufacturing demands on machining

Process Control

Statistic control of machine and process capability is necessary. The capability of all check dimensions according to drawing has to be guaranteed. The check dimensions (Cpk≥1.33) have to be monitored and evaluated during production. Check dimensions have to be fixed in an inspection chart and have to be shown on demand any time. Data has to be stored for at least 3 years.

3. Additional Manufacturing Substances

Following lubricants and coolants are acceptable without further approval:

Oil: ARAL Sulnit HM – see specification 3170, status 03.12.99

Emulsion: ARAL Sarol 471 EP ca 15% in tap water, status 01.12.99 Castrol Alusol XT FF, status 12.02.2009

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防腐蚀乳状液：Dansoclean KE8070，状态 10.03.2009。

以下所涉及的清洗液无须额外许可。

SurTec131-见 SurTec 规格，状态 03.03.99

SurTec086-见 SurTec 规格，状态 08.11.00

禁止使用含硅的物质，其余物质的使用须得到 UAES 的认可。以上要求适用于所有铝壳体的生产过程。并且同时满足博世标准 N2580。

Corrosion protection emulsion: Dansoclean KE 8070 – see specification, status 10.03.2009

Following substances for the washing process are acceptable without further approval:

SurTec 131 – see specification SurTec, status 03.03.99

SurTec 086 – see specification SurTec, status 08.11.00

Do not use any other substances. For other substances the agreement of the UAES is necessary. It applies to all additional manufacturing substances the prohibition of silicone and compliance with the Robert Bosch Norm 2580.

法规与标准要求

2000/53/EC 汽车类废品指令 欧盟议会和欧盟理事会 2000 年 9 月 18 日第 2000/53/EC 号

N2580-1 BOSCH 物质的禁用及申报标准

注：以上要求按照法规与标准的最新版本。

Regulations and standard requirements

2000/53/EC Directive 2000/53/EC of the European Parliament and of the council of 18 September 2000 on end-of life vehicles N2580-1 Prohibition and declaration of substance

Note: All above requirements in accordance with regulations and standards to the latest version.

4 清洗过程要求

质量要求：零件须是干态，无斑点且无毛刺。

尘埃物质的残留：尘埃物质的残留必须定期以文档形式定义（残留油脂（ mg）,残留灰尘（ mg））

最大残留油脂量： 6-15mg;

最大灰尘残留量： 3 mg；

4. Manufacturing demands on washing process

Quality requirements: Parts must be dry and free from borings and burrs.

Remaining dirt : The amount of remaining dirt has to be determined and documented regularly (amount of remaining oil (mg)/ amount of remaining dirt (mg)).

Maximum amount of remaining oil: 6 - 15 mg

Maximum amount of remaining dirt: 3 mg

机加壳体的最大存储时间：

若存储时间超过三周，铝壳体在装配之前须重新进行清洗。

Maximum storage periods for machined housing without Dansoclean KE 8070 :

If storage period exceeded 3 weeks, then the housing must be washed again before assembly.

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5 机加壳体的终检验

5.1 目检

根据如下规范，清洗工艺之后，零件必须 100%目检。 不合格的零件必须选出。

5.1.1 毛刺

以下标注出的孔必须检测毛刺：

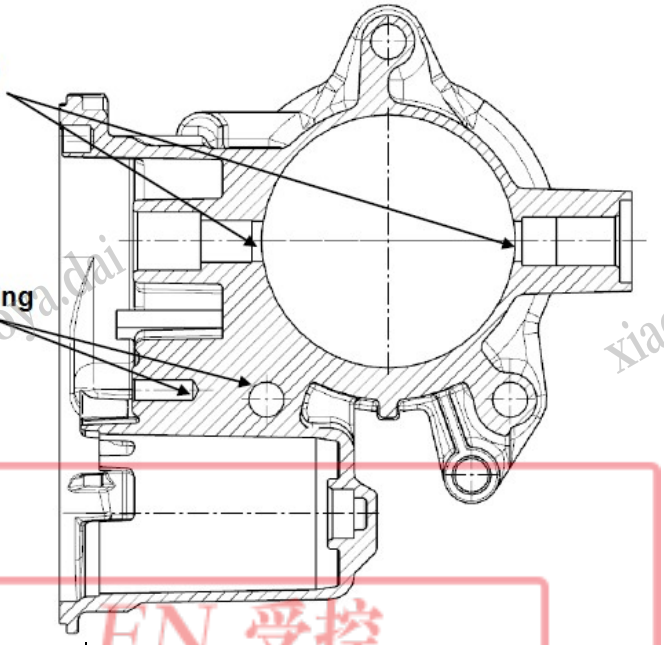
5. Final Tests on machined Throttle Housing

5.1 Visual Tests

100% of the parts have to be checked after the washing process according to the following specifications. Parts which are not perfect have to be sorted out.

5.1.1 Burrs

Following bores have to be checked for burrs:



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表面：

孔和表面必须完全清理及处理。不得存在擦痕和损伤，标明的倒角必须存在。分模线及浇口须根据毛坯图纸。毛坯孔处不得出现铸造残留。由于压铸制造工艺的不同导致出现的色差是允许的。

由于修模导致的铝壳体表面结构变化是允许的，最大 0.4mm。

压铸模尺寸超差导致的修模是允许的。

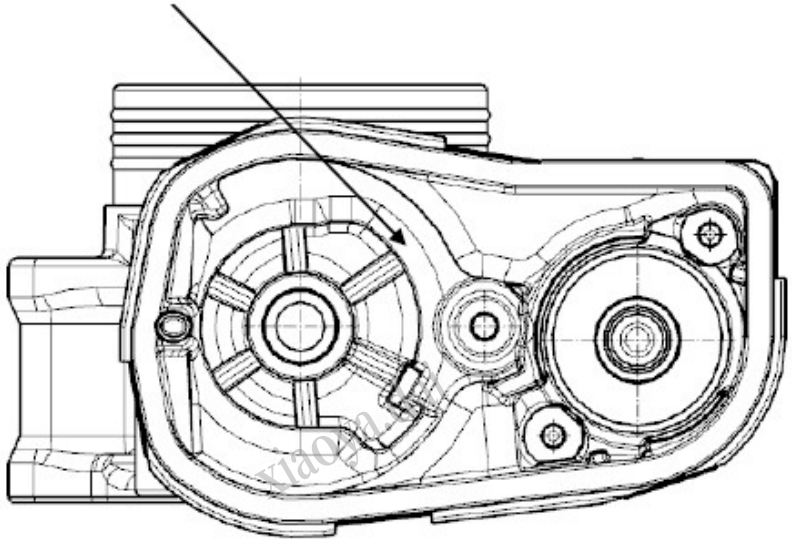
Surfaces

Bores and surfaces have to be clean and processed completely. Chatter marks and damages are not permissible, all chamfers must exist. Mold release edge and dead head gate must be according to drawing for die casting part. In bores of rough part residuals of casting are not allowed. Color variations and different optical surface structure from exfoliations, which appear from the manufacturing method in the die cast, are allowed. Change of the surface structure on the die cast housing, due to repair and rework up of the tool are permitted to max. 0.4mm. Rework of the die cast mold within the tolerance limits are allowed.

5.1.2 毛刺和废屑

5.1.2 Borings

Flashes and stucked borings in the gear box are not allowed.
在齿轮腔体中不允许有毛刺和卡住的金属屑



5.1.3 气孔及缩孔

气孔尺寸≤Φ0.7mm 是允许的，但不得相连。要求如下：①

在气道机加工区域，最多允许出现两个 Φ1.2mm 的气孔。如下图所示。①

5.1.3 Pores and Shrink holes

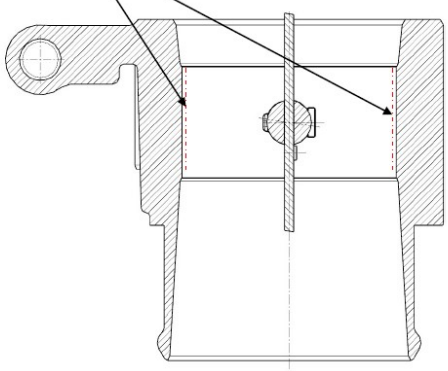
Pores Ø 0.7mm are allowed. But no chain formation. Following regulations are valid: ①

Throttle Bore:

In the area of the machined bore area maximal 2 pores up to Ø 1.2mm are allowed, as showed in following picture. ①

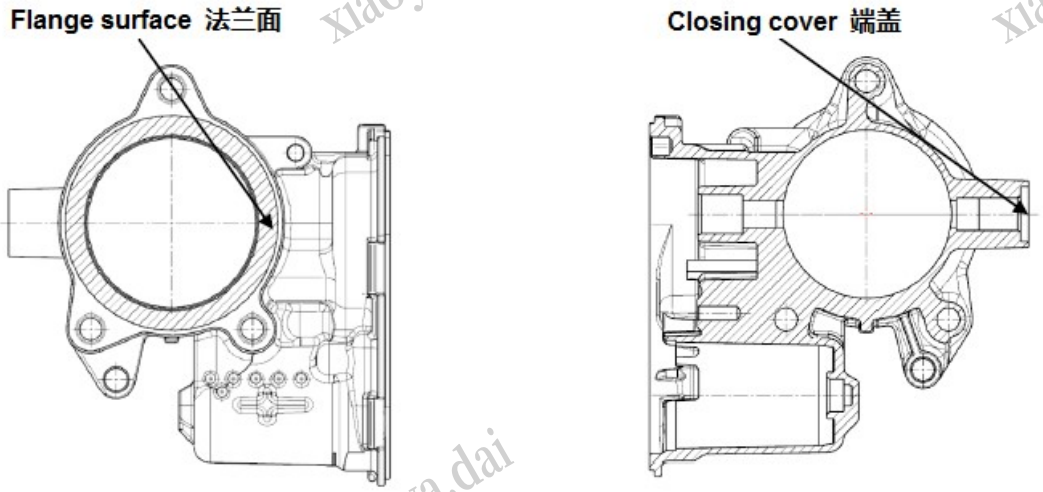
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Outside of this area maximum 5 pores with maximum Ø 1.2mm are allowed.
在此区域之外，最多允许5个不超过直径1.2mm的孔



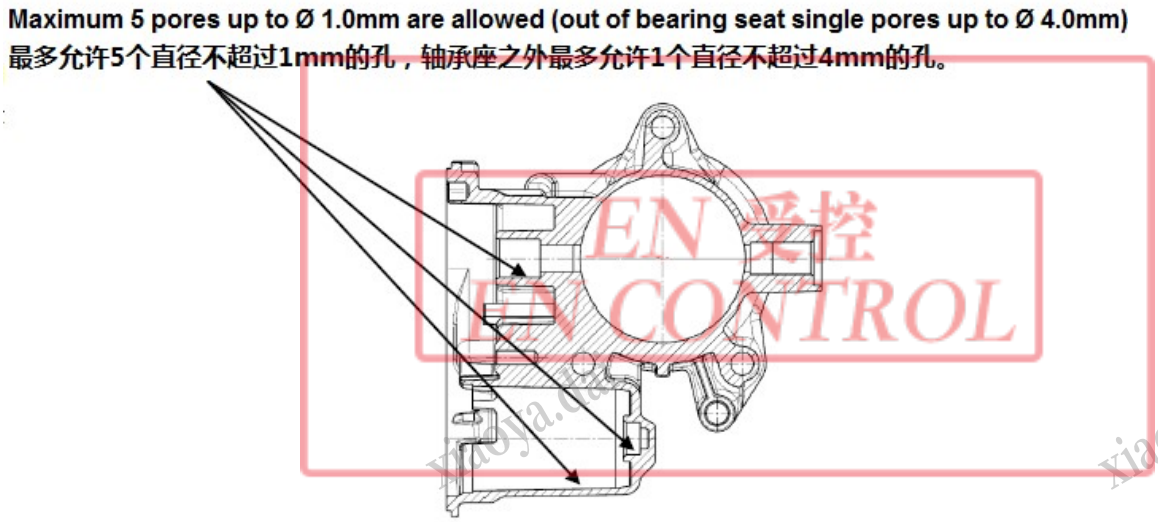
法兰面，端盖位置：①
最大允许 Φ0.7mm 的孔。但不允许在密封面上出现相邻的缩孔。

Flange Surface, Closing Cover: ①
Maximum pores up to Ø 0.7mm are allowed.
Connection of pores in direction of sealing



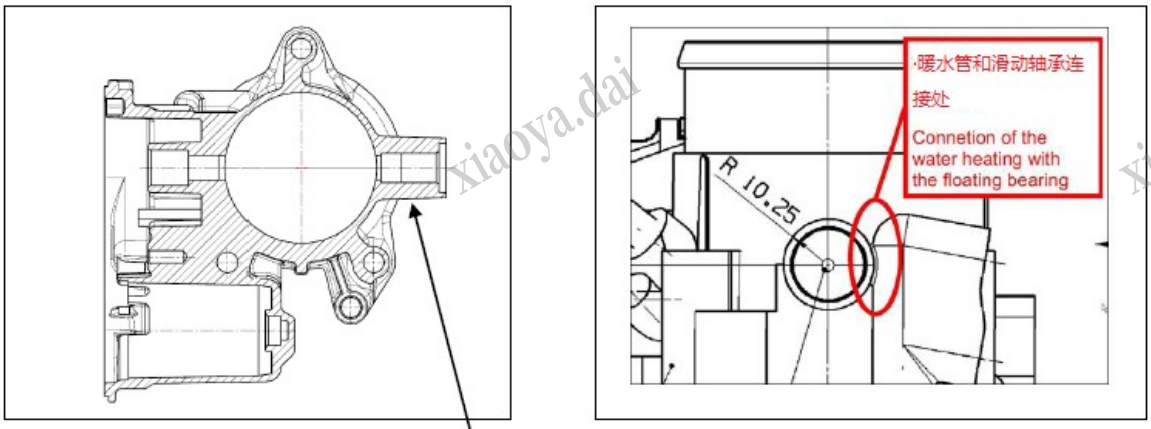
轴承安装面（阀轴，中间齿轮栓，电机轴承）

Surfaces of Bearing Seats (Shaft, Bearing Bolt, Motor Bearing):



在有加热水管出现的滑动轴承侧：

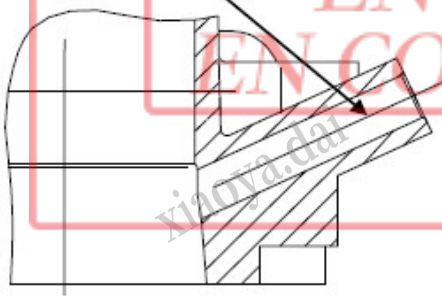
Surfaces of Bearing Seats for the slide Bearing:



With connection of the water heating with the floating bearing:
Maximum 5 pores up to Ø 1.0mm are allowed (out of bearing seat single pores up to Ø 4.0mm)
暖水管和滑动轴承连接处
最多允许5个直径不超过1mm的气孔（轴承座之外最多允许一个直径不超过4mm的孔）
Without connection of the water heating with the floating bearing:
Maximum 3 pores up to Ø 1.0mm are allowed Maximum 2 pores up to Ø 3.0mm are allowed Length of connections up to 3mm (out of bearing seat single pores up to Ø 4.0mm)
在非暖水管和滑动轴承连接处
最多允许3个不超过1mm的孔，最多允许2个不超过3mm的孔，连接处最长3mm（轴承座之外最多允许一个直径：不超过4mm的孔）

对于正压力和负压力存在的连接位置：

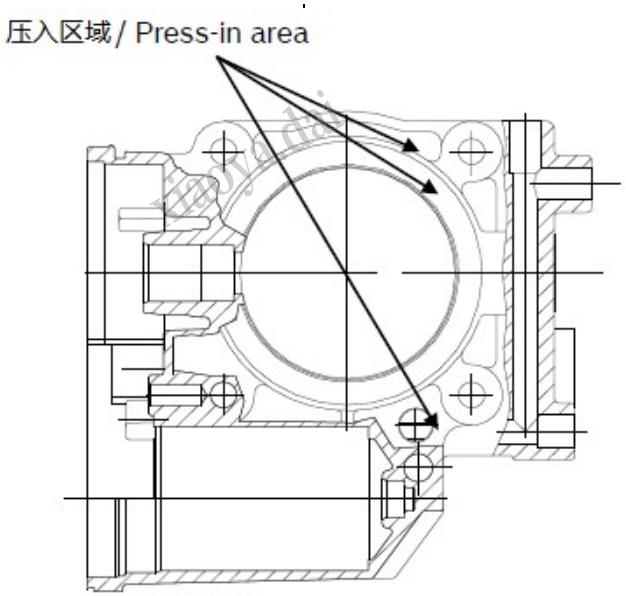
Maximum 5 pores up to Ø 1.0mm are allowed.最多允许5个直径不超过1mm的孔



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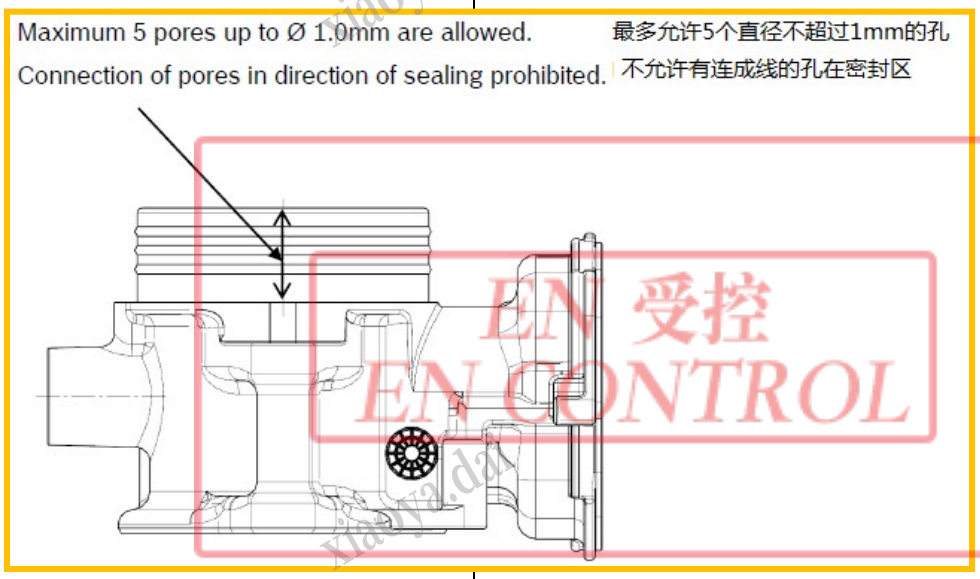
对于加热水管部位：
在水管压装区域（至 2 倍直径区域）最大孔径为Φ1.0mm 的孔最多出现 5 个。
对已出现的孔，泄露测试是判断合格与否的标准。

In press-in area (up to 2 x diameter) maximum 5 pores up to Ø 1.0mm are allowed.
The leakage test is a decisive criterion for the remaining bore.



Air Scoop: ①

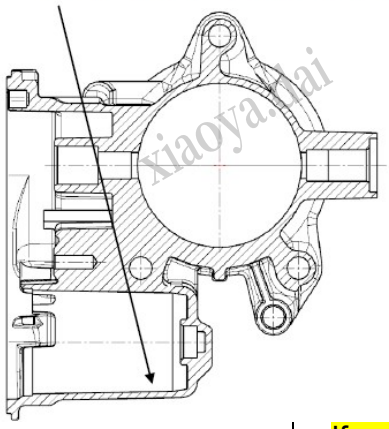
进气软管处：①



电机腔体处：

Motor Bore：

Single pores up to Ø 4mm are allowed. 最多允许一个直径不超过4mm的孔



如有客户特殊需求的（如 VW/…），按照客户要求执行。①

If customer has special requirement(eg VW…), follow customer requirement. ①

5.2 水管处的泄露测试（如果有水管）

产品必须用压降法进行 100%泄露测试，合格产品须做永久标记。

测试条件：

测试压力（绝对压力）：Pabs=3.5bar

最大流量率： ΔV≤4.0cm³/min

测试温度： T=24°C

5.2 Leakage Test Water Heating(if existing)

Leakage tests have to be done as a 100% inspection according to the method of pressure drop. Perfect parts have to be marked permanently.

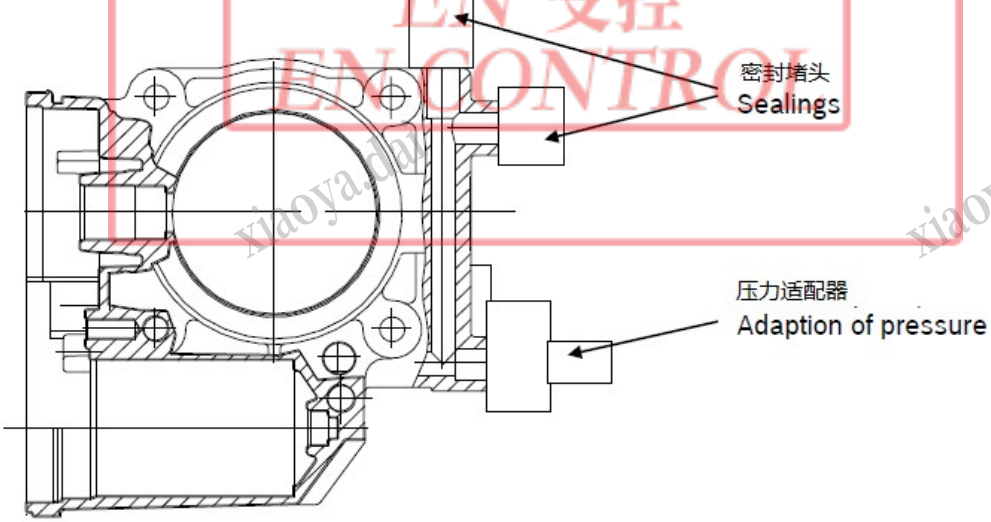
Test conditions:

Test pressure, absolute: pabs = 3,5 bar

Maximum volume flow rate: Δv 4,0 cm³/min

Test temperature: T = 24°C

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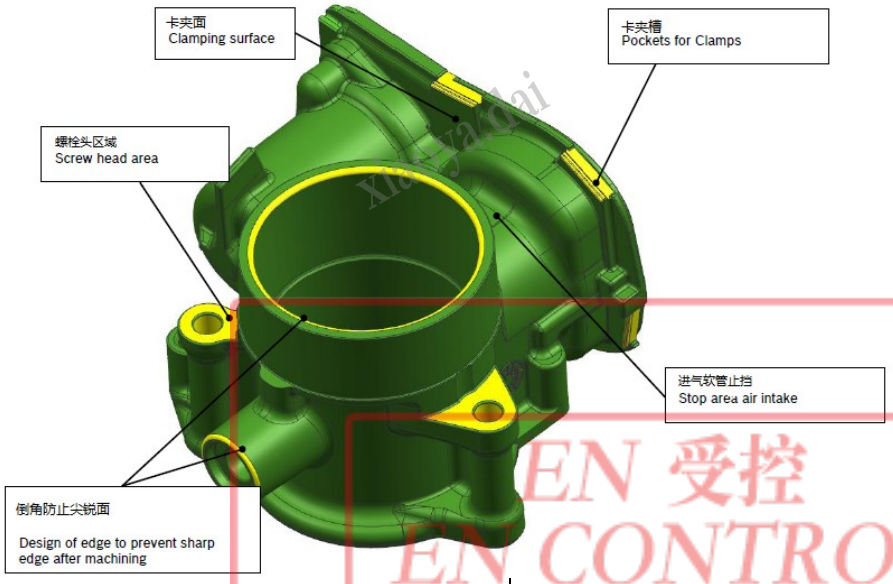
6 允许的热裂纹

6 Permissible Heat Cracks

颜色 Color	热裂纹最长允许长度 Maximum high for heat cracks in mm
grün 绿色	0,8
gelb 黄色	0,4
rot 红色	0,2

上表中的三个区域颜色区分，在下图中分颜色体现。非绿色区域的功能已明确指示出。

In the following pictures the 3 areas that are mentioned in the above table are shown with colors. For all areas that are not green the function is given.



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图3 / **Figure 3:**

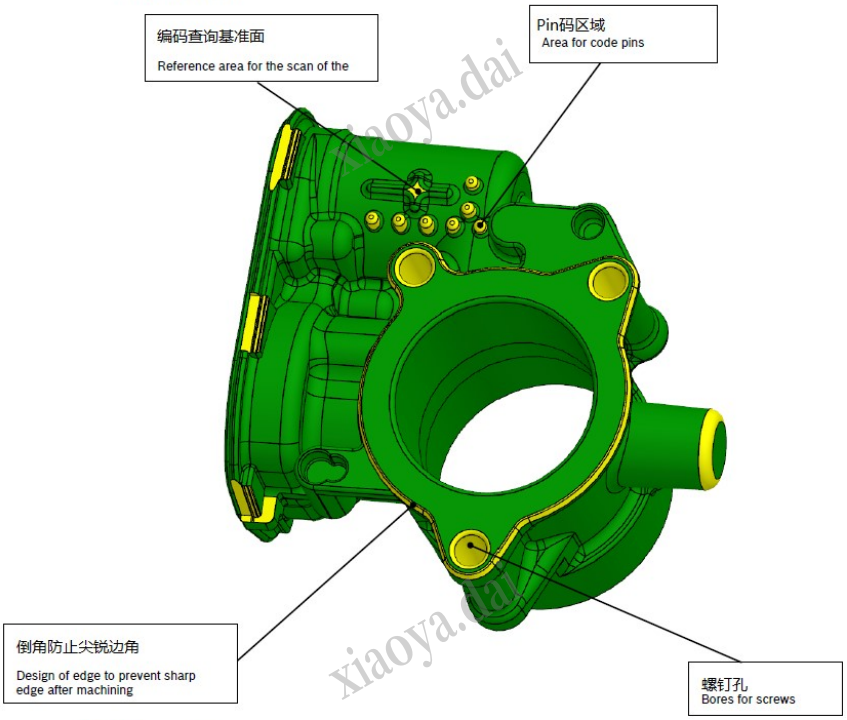
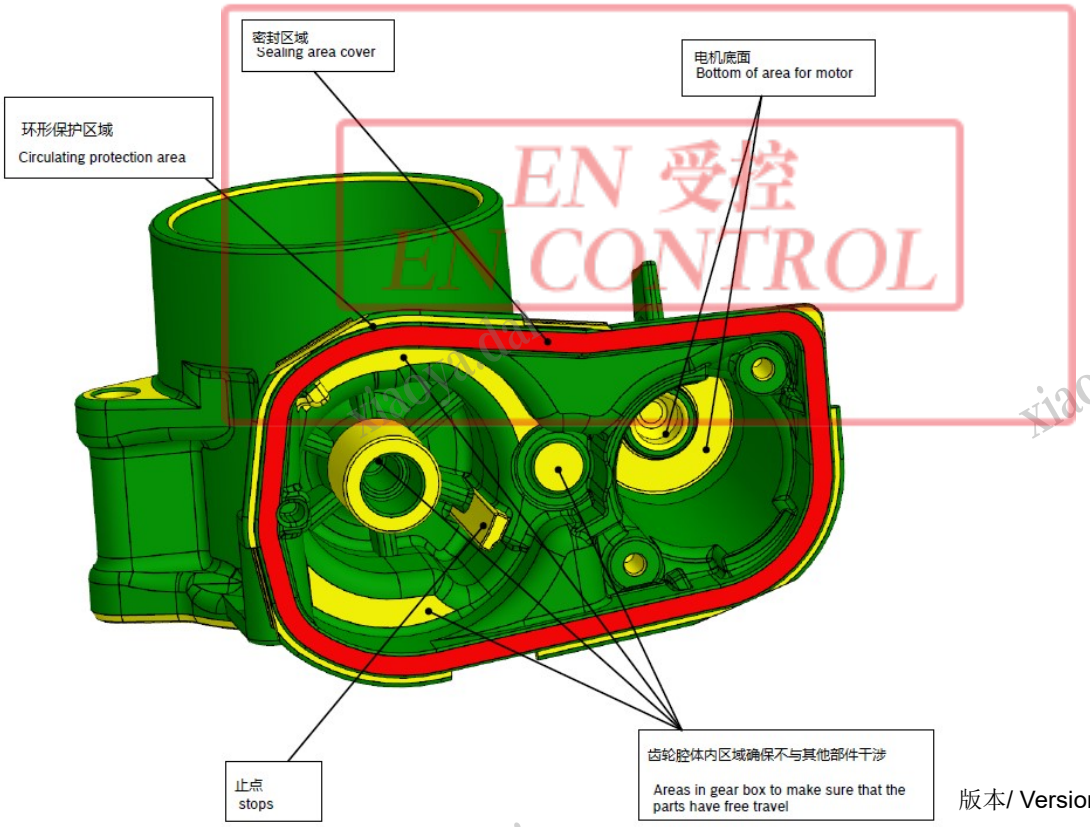


图2 / **Figure 2:**



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备注：

尖锐的，大的热裂纹须通过喷砂去除。

如有必要，可以定义允许的热裂纹大小。该状态由供应商和 UAES 质量部门共同确定。

对于壁厚较薄处（不大于 1.7mm）的热裂纹高度应不大于 0.2mm。①

进气软管接口处不允许热裂纹。①

7 定义 H 面的测量点

下图中的 12 个测量点用来确定 H 面，即铝壳体和盖板之间的密封面。

Remark:

Sharp, big heat cracks have to be eased through the sand blasting.

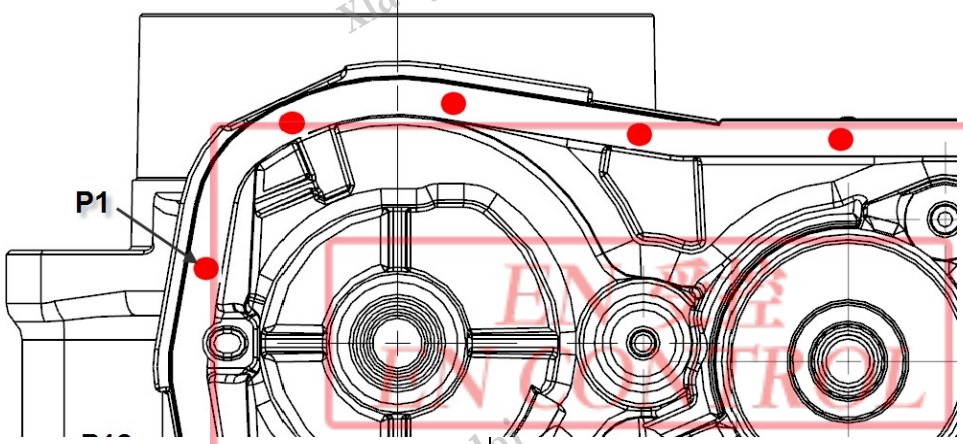
If necessary there will be defined samples that show the sharpest heat cracks that are allowed. The parts will be defined between supplier and UAES quality department.

If the wall thickness is less than 1.7mm，the high of heat crack should be no more than 0.2mm. ①

On the sealing surface of air Scoop, the heat crack is not allowed. ①

7 Measuring-points to define the plane H

The 12 measuring points shown in Figure below have to be used for the determination of the plane H /sealing surface of the cover/lid).



8 定义铝壳体密封面的平面度测量

测量铝壳体密封面

PPAP 测量：密封面每间隔 2mm 取点测量，即~282mm=141 个点

PPAP 测量时，每穴测 25 件。

批产测量：密封面每间隔 5mm 取点测量，即~282mm=56 个点

批产时平面度每班须进行首末件测量。

8 Definition flatness measurement Throttle Housing (cover sealing surface)

Measurement throttle housing sealing surface

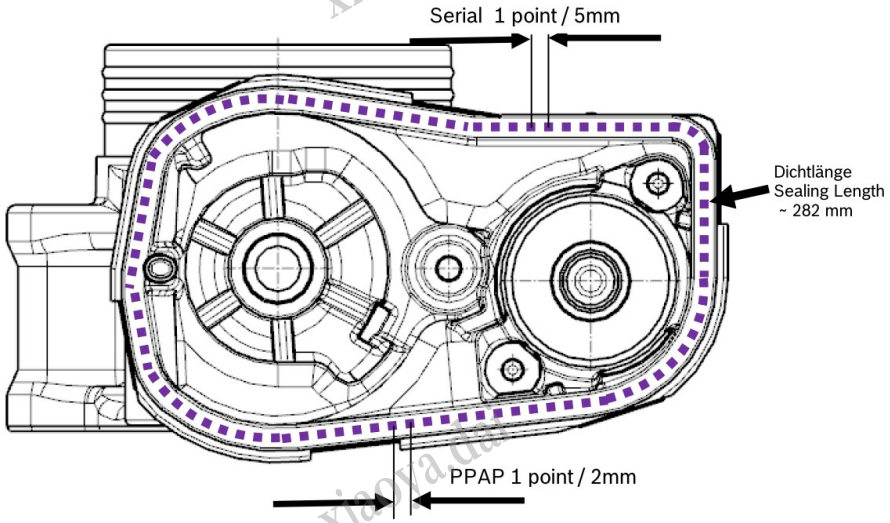
PPAP measurement：

1 point per 2mm sealing surface throttle housing ~282mm=141 points

The PPAP testing is conducted with 25 parts per cavity.

During serial production:

1 point per 5mm sealing surface throttle housing ~282mm=56 points
The serial production test is performed per shift, and at the start of production and end with one part per cavity.



9 毛坯壳体的基准及方向

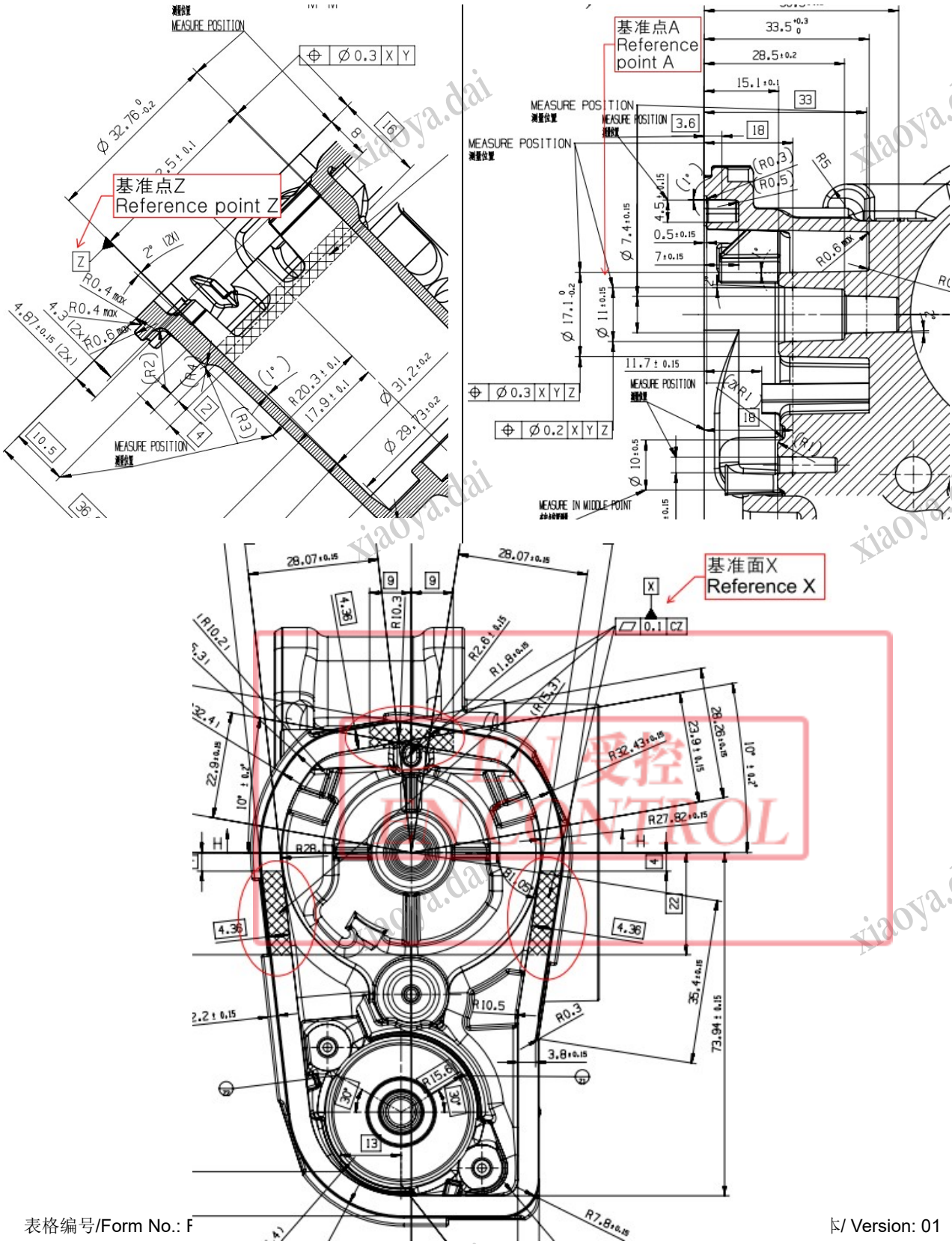
空间方向	H基准
水平面基准	A基准(ø11,0 +/-0,15) Z基准 (ø31,60)
原点	基准点A在基准面X上的投影
基准的转动	角度的为 25,36/30,55 和 2,22 (2,2727°)

9 Orientation of the not machined throttle housing

space direction	reference H
orientation of the plane	Reference A(ø11,00+/-0,15)/ reference Z (ø31,60)
zero point	projected reference A on reference X
rotation of the orientation	angle of the dimension 25,36/30,55 and 2,22 (2,2727°)

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UAES	订 货 规 程 **Order Specification**	编号：F 01R L0Y 033 No.: F 01R L0Y 033 版本：02 Version: 编制日期：220505 Issued Date: 页数：第15页 共 17 页 Page of



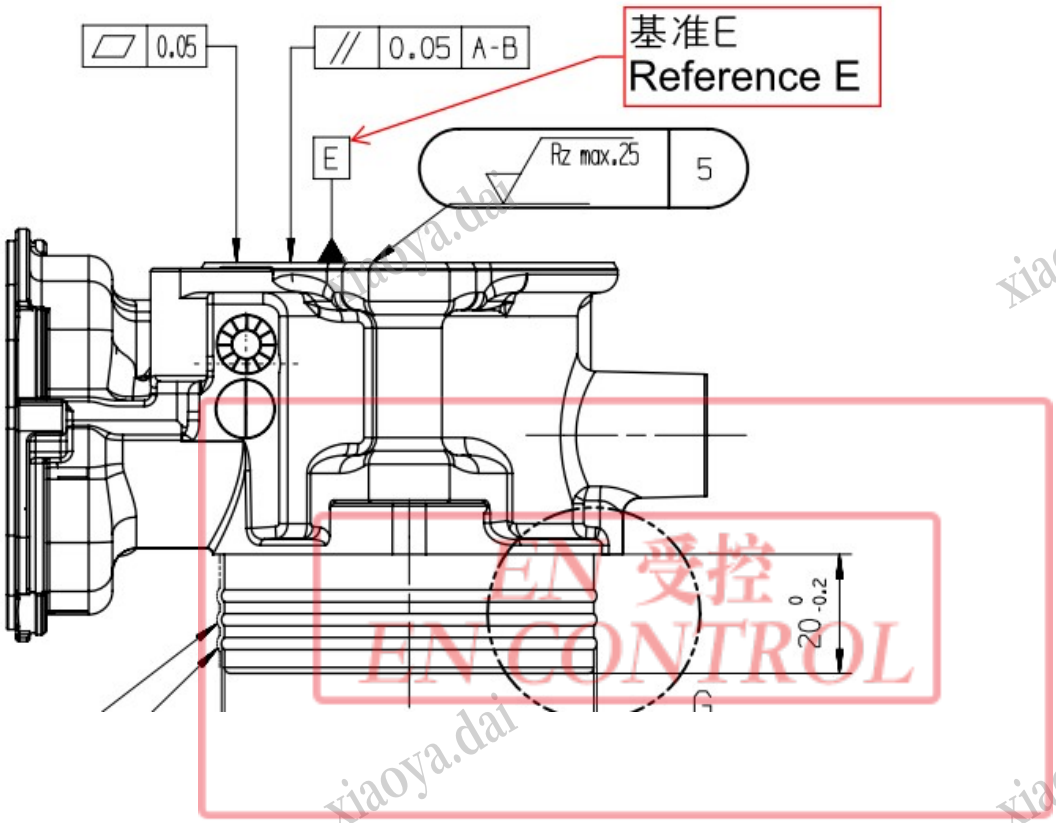
LIU 受控 CONTROL

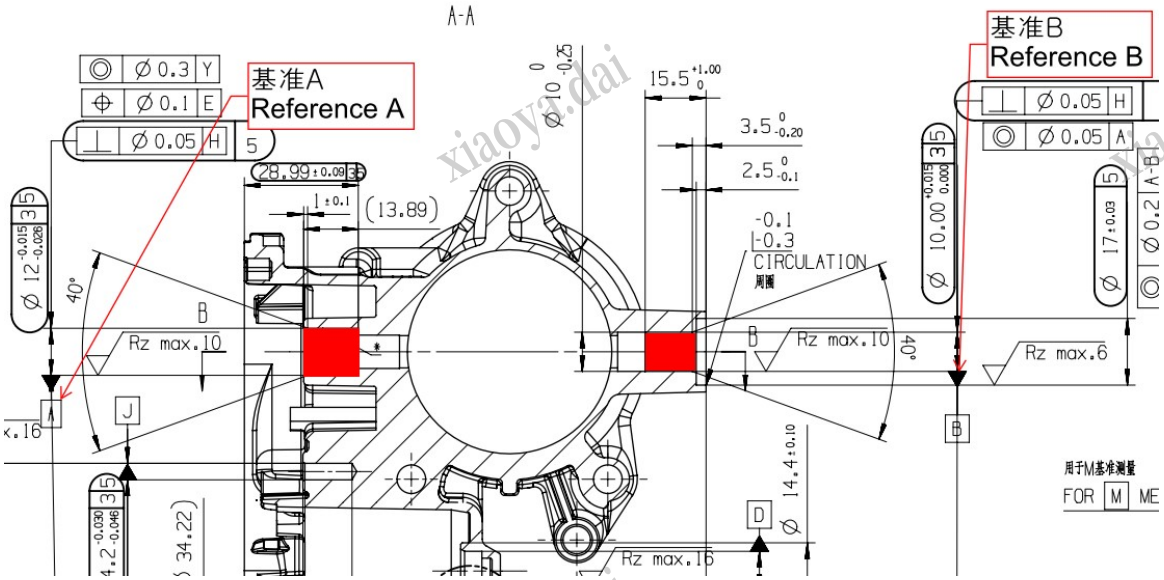
10 机加工壳体的基准及方向

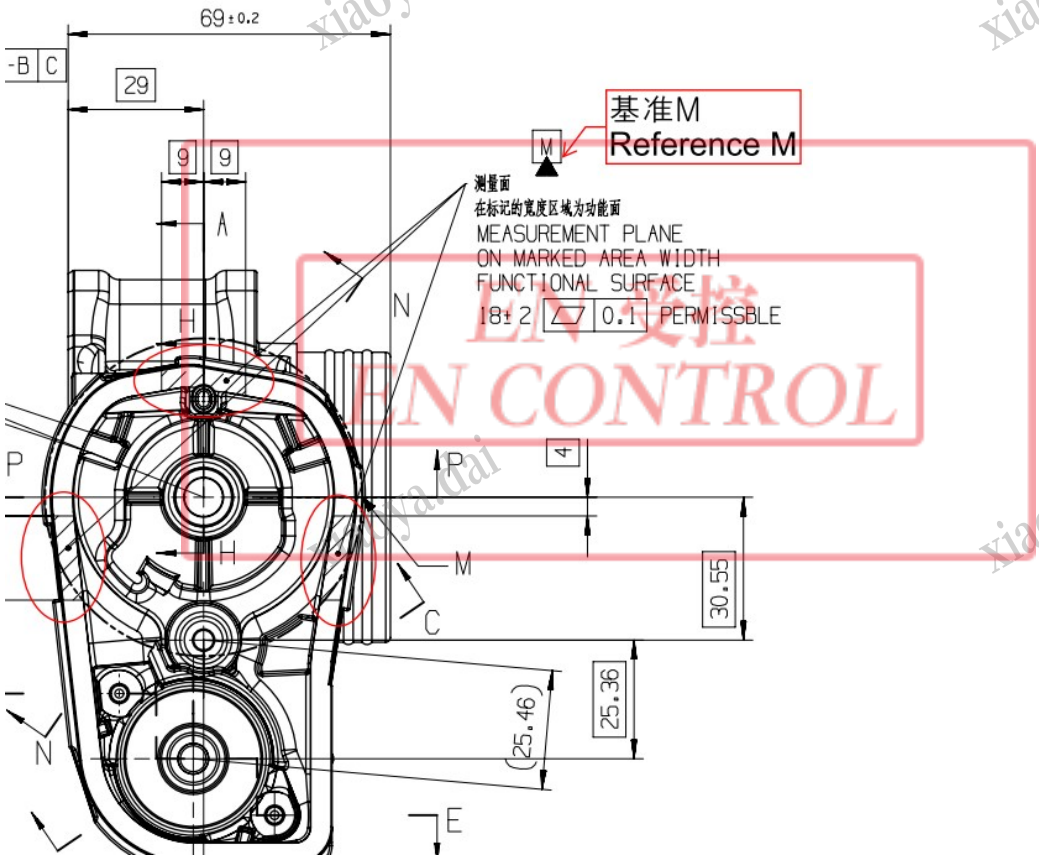
方向	E基准
水平面基准	A-B基准的连线
原点	A-B基准的连线和M基准的交点
基准的转动	无

10 Orientation of the machined throttle housing

space direction	reference E
orientation of the plane	connection between reference A / reference B
zero point	Intersection between A-B and reference M
rotation of the orientation	not necessary







EN 受控
EN CONTROL